

## 1. From Algorithms to Chaos: The Nature of Unpredictability [100 points]

### The "Homebrew" Random Number Generator

- Write a Python function (or class) `my_lcg(n, seed, a, c, m)` that generates a sequence of  $n$  numbers using a Linear Congruential Generator (LCG). The raw output will be integers between 0 and  $m - 1$ . Return the array divided by  $m$ , so the values are floats in the range  $[0, 1)$ .
- Use your function to generate two datasets of  $N = 2000$  points each: The "Bad" Generator:  $a = 1229$ ,  $c = 1$ ,  $m = 2048$ . The "Standard" Generator (MINSTD):  $a = 16807$ ,  $c = 0$ ,  $m = 2^{31} - 1$ . The latter one was used by Apple and C++ libraries in the 90s.
- Return map: Create a figure with 3 subplots side-by-side. Generate scatter plots ( $x_n$  on the x-axis,  $x_{n+1}$  on the y-axis) for: Your "Bad" LCG data, your "Standard" MINSTD data, and Python's built-in `numpy.random.uniform` (The Mersenne Twister algorithm). Inspect the "Bad" generator plot. You should see a pattern known as "Marsaglia's Planes" (lattice structure). Why is this lattice structure dangerous for scientific simulations (e.g., simulating a gas in a 2D box)? Does the MINSTD generator show these lines? Does Numpy?

### Deterministic Chaos (The Logistic Map)

- For the Logistic Map set the growth parameter to the chaotic regime,  $r = 1.0$ . Run the simulation for 100 steps for two cases: case A:  $x_0 = 0.4$ ; case B:  $x_0 = 0.40000001$  (a change of  $10^{-8}$ ). Plot both trajectories on the same Time vs. Value graph. At what step do they diverge?
- We now compare our "Bad Randomness" (LCG), "Good Randomness" (Numpy), and "Chaos" (Logistic Map). Generate a sequence of 2000 steps using the Logistic Map ( $r = 1.0$ ). Create a Return Map scatter plot ( $x_n$  vs  $x_{n+1}$ ), just as you did in Part c).
- Compare the Return Maps of the "Bad LCG" and the "Logistic Map". Both systems produce outputs that look random in a time series. Both systems have a visible geometric structure in the Return Map. Describe the geometry and explain the difference between *Stochastic Noise* (ideal randomness) and *Deterministic Chaos* based on these visual structures.
- Calculate and plot the histogram (probability density) of: The Numpy random sequence and the Logistic Map sequence ( $r = 1$ ). Most random generators try to achieve a Uniform Distribution (flat histogram). What is the shape of the Logistic Map's distribution? (Hint: It is related to the Beta Distribution).

**The use of Generative AI is not allowed in this course!**